KERRY DEAN JR.

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years shipping production systems where mistakes have teeth: federal AI compliance pipelines, HIPAA constrained clinical platforms, satellite agtech analytics, and enterprise SaaS integrations. The last two years have been on production Large Language Model products: hybrid retrieval RAG with citation provenance, Claude Agent SDK skills, n8n agentic workflows, FHIR grade clinical AI, and a public Model Context Protocol server. The combination of regulated environment instincts and modern AI engineering is the part I bring to every project.

CORE STACK

Languages: Python, TypeScript, JavaScript, C#, Java, SQL, GraphQL

Frameworks: React, Next.js, Node.js, .NET, Entity Framework, ASP.NET, Spring

AI/LLM: Anthropic Claude API, Claude Agent SDK, OpenAI GPT-4, Gemini, LangChain, Pinecone, FAISS, RAG, hybrid retrieval, embeddings, reranking, function calling, agentic workflows, evaluation harnesses (LangSmith, Braintrust, Phoenix), Model Context Protocol

Cloud: AWS (Lambda, Bedrock, Rekognition, S3), Azure, Google Cloud, Docker, GitHub Actions, CircleCI, Firebase, Vercel

Data: Postgres, DynamoDB, BigQuery, Redis, Neo4j, SQL Server

Healthcare: FHIR, HIPAA audit logging, offline first clinical workflows

EDUCATION

University of Iowa — Iowa City, IA

Bachelor of Science, Computer Science. Iowa Hawkeyes Track and Field, Academic All Big Ten Student Athlete. — *August 2012 to May 2016*

PROFESSIONAL EXPERIENCE

Traverse Technologies — Remote

Senior Technical Lead (Contract) — *June 2025 to December 2025*

- Owned the Choice Hotels integration on Dash, designing the partner API contract, building availability search and booking confirmation, mapping rate codes and loyalty programs, and rolling out across the enterprise account base over six weeks. Decision I would defend: keep the booking confirmation flow synchronous to the partner even when fan out async would have been faster, because reservation desks need a deterministic confirmation number on the first call, not eventual consistency.
- Shipped enterprise SSO and directory sync against Okta and Azure AD on TypeScript and React with C# and .NET services, then hardened release safety with a gated GitHub Actions and Docker pipeline that closed the loop between identity events and traveler profile reconciliation. Decision I would defend: anchor identity events as the source of truth and reconcile traveler profile state on a delayed worker, because a corporate traveler whose Okta deactivation has not propagated still must not be able to book on the company account.

WellBe Senior Medical — Chicago, IL

Senior Software Engineer — *June 2023 to February 2025*

- Owned FHIR API design on Luminate exchanging eligibility, claims history, and visit documentation with payer and health plan systems, with every model invocation and PHI access path written inside a HIPAA audit log boundary. Decision I would defend: write the audit row in the same transaction as the PHI read rather than asynchronously, because the only audit trail that survives an OCR cross check is one that cannot exist without the read itself committing.
- Built the offline first tablet sync layer providers used to run standardized clinical assessments inside senior patient homes with no connectivity, syncing assessments, vitals, and chart write back to AWS when service returned, while leading a team of seven engineers shipping Luminate across .NET Entity Framework, TypeScript, Python, Firebase, and AWS. Decision I would defend: treat the tablet as the system of record between sync windows rather than a cache, because providers cannot wait for a server round trip mid visit to know whether a vitals entry actually persisted.

FwdThink — Remote

Senior Generative AI Software Engineer — *June 2024 to December 2024*

- Cut intake backlog from weeks to hours on the federal EEO complaint pipeline by pairing an AWS Rekognition OCR stage with a GPT-4 typed schema extractor that pulls complaint type, parties, alleged violations, and requested remedies, then routing each case with an LLM classifier that emits an auditable justification reviewers could accept or reject. Decision I would defend: make the model emit the justification token before the routing token, because forcing the explanation to come first changes what the model commits to, not just what the human reads.

- Compressed multi day procurement desk research to seconds by indexing contract databases, solicitation documents, and market reports into a vector store with chunk level citation provenance, then lifted faithfulness past 90 percent and cut confident hallucinations from 11 percent by adding an evaluation harness that runs in CI on every prompt or model change. Decision I would defend: gate deploys on a citation faithfulness metric rather than answer win rate, because a confidently wrong answer with no traceable source is the failure mode federal reviewers cannot recover from.

Syngenta Chicago, IL
Senior Software Engineer *May 2021 to June 2023*

- Reduced agronomist time to field insight by shipping a satellite imagery and weather overlay dashboard on interactive maps zooming from regional view down to individual field boundaries, on a Python and AWS Lambda backend with GraphQL and DynamoDB powering real time decisions for growers managing yield and risk. Decision I would defend: serve tile imagery from precomputed cloud free composites rather than recomputing per request, because agronomists need same day comparability between fields and live cloud cover would let weather drive the answer instead of crop state.
- Contributed reusable map, virtualized data grid, and chart components into Syngenta's open source React UI library now used across the organization, then wired sensor driven decisions into the analytics layer for crop yield and risk management. Decision I would defend: virtualize at the row index level instead of pixel scroll position, because crop sensor feeds change row count between renders and pixel virtualization quietly drops the sensor rows the user actually needed to see.

Argonne National Laboratory Chicago, IL
Software Developer Consultant *November 2020 to April 2021*

- Shipped the fleet electrification modeling tool and comparative TCO dashboard federal researchers and fleet managers used to build the business case for transitioning to electric vehicles, on Python ETL pipelines into Neo4j pulling EPA emissions data, fleet telemetry, and EIA carbon intensity feeds. Decision I would defend: model the fleet as a graph rather than a table, because charging infrastructure constraints and route dependencies only express cleanly when vehicles, depots, and corridors are first class nodes with weighted edges.

Enablon Chicago, IL
Software Engineer *February 2018 to October 2020*

- Modernized legacy client deployments on Enablon's sustainability and compliance platform without breaking decade old report customizations across global enterprise tenants, on React, Redux, TypeScript, Node.js, C#, AWS, and SQL Server. Decision I would defend: stand up a parallel customization translator rather than a flag day migration, because one broken sustainability report at a global enterprise customer is the kind of regression a careers worth of trust does not survive.

Group O Iowa City, IA
Software Developer *October 2016 to January 2018*

- Built B2B order routing apps on React, Redux, Angular, and Java with ASP.NET, C#, SQL Server, MongoDB, and PHP backend services, tuning application performance with targeted optimizations across customer facing surfaces. Decision I would defend: cache at the order routing decision layer rather than the data fetch layer, because the same product can route to different fulfillment centers depending on rules that change faster than the underlying inventory.

SELECTED AI PROJECTS

Claude Agent SDK Production Skills

- Authored five production Claude Agent SDK skills (agentic workflow generation, KDP publishing, Etsy API client with image generation, browser automation) on a deterministic scaffold plus LLM hybrid pattern with retry, fallback, and validation gates between every stage. Open sourced as claude-skill-template on GitHub. Decision I would defend: keep the scaffold deterministic and call the model only on judgment paths, because that is what makes cost and reliability curves predictable as the catalog grows.

n8n Agentic Workflow Library

- Built a public library of 41 agentic n8n workflow packs covering function and tool calling, RAG retrieval, hybrid vector search, and multi step agent orchestration with first class error handling and integrations across Slack, Gmail, HubSpot, Airtable, Notion, OpenAI, and Anthropic. Public at github.com/whodeanie/n8n-agentic-workflows. Decision I would defend: idempotency at the webhook ingestion layer, not the workflow boundary, is the only design that survives duplicate deliveries without producing duplicate side effects downstream.

MCP RAG Starter

- Packaged a hybrid retrieval RAG pipeline behind a Model Context Protocol server so any MCP compatible client (Claude Desktop, Cursor) can call into the same substrate. BM25 lexical search fused with dense vector via reciprocal rank fusion; citation provenance survives every transformation and resolves chunk identifiers back to source document, page, and character offsets. The evaluation harness has already caught a chunking regression that quietly halved retrieval quality. Decision I would defend: treat evaluation as a first class concern with a pre deploy gate, because the alternative is silent regression on the only axis users care about.